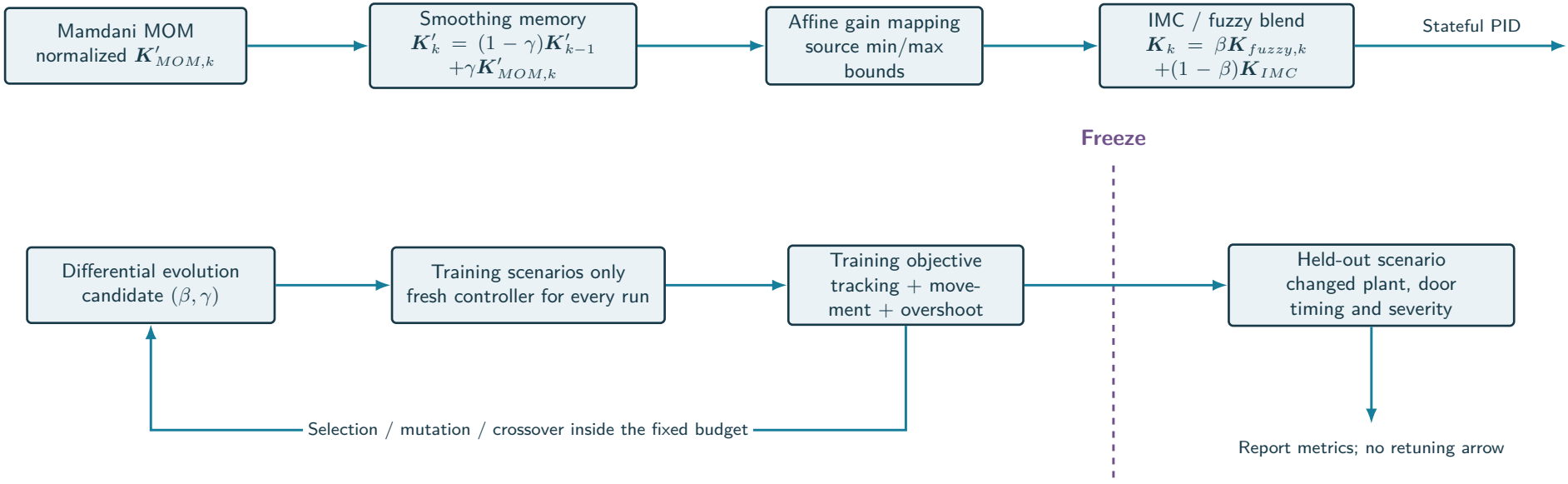
Optimization: tune on training scenarios, freeze, then evaluate

Design variables: $\beta \in [0, 1]$ and $\gamma \in [0.05, 1]$ for the optimization notebook. $\beta = 1$ uses fuzzy gains only; $\beta = 0$ uses the fixed IMC gains. Smaller γ slows gain changes. These mechanisms are post-thesis extensions.



What is held constant
Training scenarios, 3 s simulation sample period, actuator limits, random seeds and evaluation budget. Start each trajectory with a fresh controller; otherwise integral and scheduling history contaminate comparisons.

What the optimum means
Best sampled training objective under the declared search, not a global optimum or a physical performance guarantee. Preserve the evaluation count and compare alternatives under matched information and computation.

What must be reported
Independent IAE, overshoot, recovery and total variation, plus adverse outcomes. The historical 22% / 3% observations are reported source results; they are not reproduced by this synthetic search.

Workflow follows notebook 04 and FuzzyPID.step. The evaluation scenario is separated from candidate selection. Search settings, parameter distributions and the synthesized dense rule table remain documented research choices.

Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.